

Coding Challenge: Mine Site Truck Tracking System

Technical Requirements

Core Technologies

- Angular 18+ (latest stable version)
- TypeScript
- RxJS for reactive programming
- CSS

Application Features

1. Simple Map Display

- Display a **simple 2D canvas or div-based map** (1000x800px coordinate system)
- Show 2 static zones:
 - **Loading zone** (top-left, marked with distinct color)
 - **Dump zone** (bottom-right, marked with distinct color)
- No zoom/pan needed - keep it simple

2. Truck Display

- Display **1 truck** on the map
- Each truck shows:
 - **ID** (e.g., T-001)
 - **Status** with color coding:
 - LOADING (Orange)
 - HAULING (Green)
 - DUMPING (Blue)
 - IDLE (Gray)
 - **Speed** (km/h)
- Represent trucks as colored circles or simple SVG icons.

3. Real-Time Simulation

- Use **RxJS interval** to update truck data every 2 seconds
- Generate random position changes (trucks should generally move between loading and dump zones)
- Randomly change status based on position
- Simple position updates - no complex pathfinding needed